Apache Spark

Q: How would you handle skewed data in a Spark job to prevent performance issues?
A:

- Use **salting** by adding random prefixes to keys to redistribute data evenly.
- Employ **broadcast joins** when one dataset is small enough to fit in memory.
- Partition the data strategically using custom partitioners.
- Leverage spark.sql.autoBroadcastJoinThreshold for automatic optimization.

Q: What is the difference between Spark Session and Spark Context? When should each be used?
A:

- **Spark Context**: Manages the cluster connection and is the entry point for RDD-based APIs.
- **Spark Session**: A unified entry point for managing Spark applications, including Spark SQL, DataFrames, and more.
- Use Spark Session for new applications as it consolidates functionalities of SparkContext and other contexts.

Q: How do you handle backpressure in Spark Streaming applications?
A:

- Enable **backpressure** using spark.streaming.backpressure.enabled.
- Configure spark.streaming.kafka.maxRatePerPartition for rate control.
- Optimize batch intervals and use checkpointing for reliability.

Apache Kafka

Q: How do you handle exactly-once semantics in Kafka Streams?
A:

- Use Kafka Streams' **idempotent producers** and **transactional APIs** to ensure atomic writes.
- Challenges include configuring proper isolation levels and ensuring consistent state management in distributed systems.

Q: What is the role of ZooKeeper in Kafka, and what are the implications of moving to KRaft?
A:

- **ZooKeeper** manages metadata, leader election, and cluster state.
- **KRaft (Kafka Raft)** eliminates ZooKeeper, offering better scalability and simpler architecture.

Q: How do you manage data retention and deletion policies in Kafka?
A:

- Configure log.retention.hours for time-based retention.
- Use log.retention.bytes for size-based retention.
- Set delete.retention.ms for log compaction to retain deleted messages temporarily.

Apache Airflow

Q: What is an Airflow XCom, and how would you use it?
A:

- XComs (cross-communications) allow data sharing between tasks in a DAG.
- Use xcom_push() to send data and xcom_pull() to retrieve it in subsequent tasks.

Q: How can you set up task-level retries and backoff strategies in Airflow?
A:

- Configure retries and retry_delay parameters in the task.
- Use retry_exponential_backoff=True for increasing retry intervals.

Q: How do you use the Airflow REST API to trigger DAGs?
A:

- Use the POST /dags/<dag_id>/dagRuns endpoint to trigger DAGs.
- The GET /dags/<dag_id>/dagRuns endpoint retrieves DAG status.

Data Warehousing

Q: How do you optimize join operations in a data warehouse?
A:

- Use indexed columns for join keys.
- Partition large datasets appropriately.
- Pre-aggregate or denormalize tables when possible.

Q: What is a Slowly Changing Dimension (SCD)? How is it implemented?
A:

- SCD handles changes in dimension data over time.
    - **Type 1**: Overwrite data.
    - **Type 2**: Add new rows for changes.
    - **Type 3**: Add columns for changes.

Q: Why are surrogate keys preferred over natural keys in data warehouses?
A:

- Surrogate keys ensure uniformity and are immutable.
- They handle changes in business logic better than natural keys.

CI/CD

Q: What are blue-green deployments, and how are they used for ETL jobs?
A:

- Maintain two environments (blue and green); one is live, the other is a staging area for updates.
- Switch to the new environment after testing, ensuring minimal downtime.

Q: How do you implement rollback mechanisms in CI/CD pipelines?
A:

- Use versioned artifacts and maintain backup databases.
- Implement feature toggles to switch off failing features quickly.

Q: How do you handle schema evolution in data pipelines?
A:

- Use schema registries to manage schema versions.
- Implement backward-compatible changes.

SQL

Q: How do you calculate a cumulative sum in SQL?
A:

```
SELECT
  category,
  amount,
  SUM(amount) OVER (PARTITION BY category ORDER BY date) AS cumulative_sum
FROM sales;
```

Q: How do window functions differ from aggregate functions?
A:

- Window functions operate on a subset of rows while retaining individual row details.
- Aggregate functions collapse rows into a single result.

Q: How do you remove duplicates in SQL without temporary tables?
A:

```sql
DELETE FROM my_table
WHERE id NOT IN (
  SELECT MIN(id)
  FROM my_table
  GROUP BY unique_column
);
```

Python

Q: How do you process large files in Python efficiently?
A:

- Use generators and the with open() pattern for memory-efficient file handling.
- Libraries like pandas can be used with chunking (chunksize) for large datasets.

Q: What are Python decorators, and how are they useful in ETL?
A:

- Decorators modify the behavior of functions or methods.
- Use them for logging, retries, or monitoring ETL processes.

Q: How do you use Python's logging module for error logs?
A:

```python
import logging
logging.basicConfig(filename='etl.log', level=logging.ERROR)
logging.error('An error occurred')
```

Azure Databricks

Q: How do you configure cluster autoscaling in Databricks?
A:

- Enable autoscaling in cluster settings by setting min and max nodes.
- Use it for handling variable workloads dynamically.

Q: How do you implement data versioning in Delta Lake tables?
A:

- Leverage Delta Lake's built-in versioning by using TIME TRAVEL queries.

Q: How do you monitor and optimize job performance in Databricks?
A:

- Use the **Ganglia UI** and **Spark UI** for metrics.

- Optimize by tuning partitions and caching frequently used datasets.

Azure Data Factory

Q: What are tumbling window triggers in Azure Data Factory?
A:

- Tumbling windows trigger pipelines at fixed intervals.
- Configure intervals and dependencies for time-bound workflows.

Q: How do you enable managed identity-based authentication in ADF?
A:

- Assign a managed identity to the ADF instance.
- Configure linked services to use the managed identity for authentication.

Q: How do you create custom activity logs in ADF?
A:

- Use the Azure Monitor integration for pipeline monitoring.
- Include logging within custom activities using Python or .NET SDKs.

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